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Page No.:
Date:

Analysis of Algorithms

* Algorithm

(1) An Algorithm is a meaningful set of instructions which is in a sequencial manner.

(2) Algorithm should satisfy the following criteria
   a) Input
   b) Output
   c) Definitness
   d) Finiteness
   e) Effectiveness

(3) Performance Analysis have 2 types
   1. Time Complexity:
   2. Space Complexity

(4)

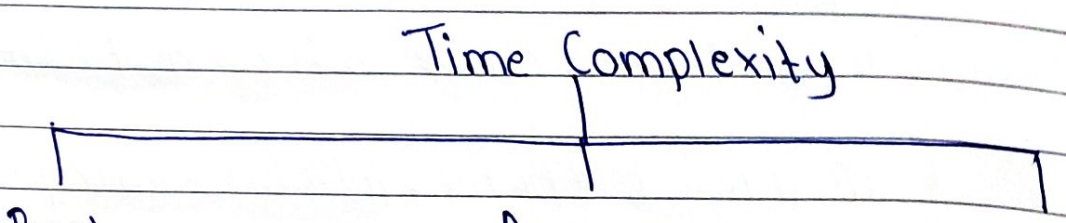
a) Time Complexity: — Best case / Worst case / Average case.
   How much time the program takes to execute

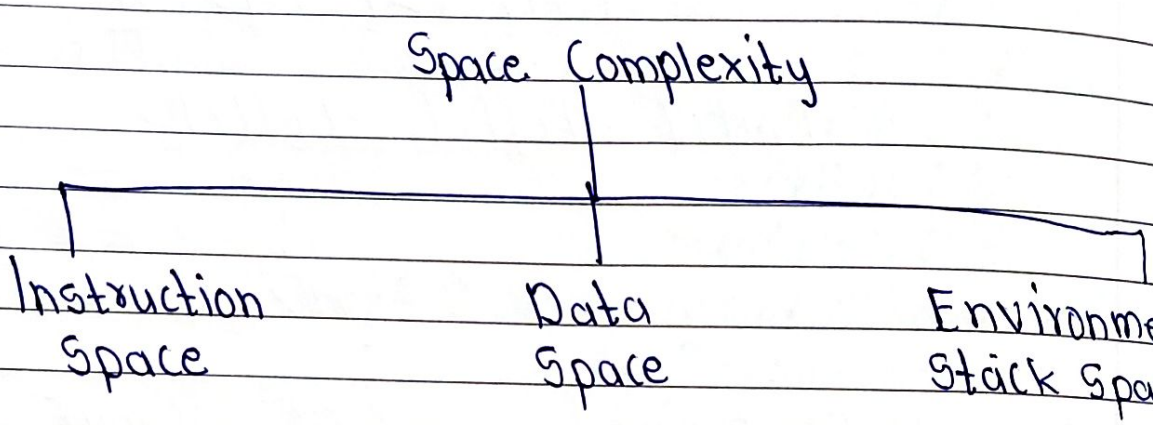
b) Space Complexity — Instruction Space / Data space / Environment stack space.
   How much space/memory a program takes.

(4) We can Compare Algorithms based on 3 ~~crite~~ criteria
   a) Execution time
   b) Number of statements executed
   c) Running time Analysis

*We always compare two algo by its worst case

Page No.:
Date:





(1) Instruction Space:-
Store the compiled version of the program instructions.

(2) Data Space:-
Store all constant and variable values

(3) Environment Stack space:-
Save information needed to resume execu-tion of partially completed functions

There are two approaches to calculate Time / Space Complexity:
1) Frequency count / Step count Method (Independent)
2) Asymptotic Notations (dependent)

* (1) Frequency count / Step count Method Rules.

(1) Ignore lower order exponents and also ignore the constant value of higher order exponent
(2) For comments, declaration
count = 0
(3) return and assignment statement
count = 1

eg ignore constant
$f(n) = 6n^3 + 10n^2 + 15n + 3$ → ignore lower order exponents.

So $f(n) = O(n^3)$

---

(2) eg: Sum of n values of an array

(1) Algorithm sum(int a[], int n)

```
S=0;
for(i=0; i<n; i++)
{
    S=S+a[i];
}
return S;
```

Time Complexity
1
n+1
n
1
2n+3
f(n) = O(n)

Space Complexity
a[] = n words
n = 1 word
S = 1 word
i = 1 word
n+3
& O(n)

2 Addition of two square Matrices of dimension

```
Algorithm addMat (int a[][], int b[][])
{
  int c[][];
  for (i=0; i<n; i++)
  {
    for (j=0; j<n; j++)
    {
      c[i][j] = a[i][j] + b[i][j]
    }
  }
}
```

Time Complexity	Space Complexity
n+1	a[][] = n^2 words
n x (n+1)	b[][] = n^2 words
n x n	c[][] = n^2 words
n+1+n^2+n+n^2	i = 1 word
$2n^2$+2n+1	j = 1 word
f(n) = $O(n^2)$	n = 1 word
	$3n^2$+3
	$O(n^2)$

Page No :
Date :

loops (By Step-count Method)

1. `for(i=0; i<n; i++)` & `for(i=n; i>n; i--)`

```
{
   statements;
}
```

```
{
   statements;
}
```

* Time Complexity of Both

→ n+1
→ n

∴ f(n) = 2n+1
f(n) = O(n)

---

2. `for(i=1; i<n; i=i+2)`

```
{
   statements;
}
```

* Time Complexity

→ n+1
→ n/2

∴ f(n) = 3n/2 + 1
f(n) = O(n)

3.
```
for (i=0; i<n; i++)
{
    for (j=0; j<n; j++)
    {
    Statements;
    }
}
```

* Time Complexity

→ n+1
→ n(n+1)
→ n×n

$f(n) = 2n^2 + 2n + 1$
$f(n) = O(n^2)$

Page No.:
Date:

loops (By tracing)

①
```
for(i=0; i<n; i++)
{
  for (j=0; j<i; j++)
  {
    Statements;
  }
}
```

Time Complexity

i	j	Statements
0	0	0
1	0	1
	1	2
2	0	3
	1	4
	2	5
3	0	6
	1	7
	2	8
	3	9
....	...	...
N	0 to n-1	n

$1 + 2 + 3 + 4 + \cdots + n = n(n+1)/2$

$T(n) = 1 + 2 + 3 + 4 + \cdots + n-1 = \frac{(n-1)(n)}{2}$

$= O(n^2)$

(2)

```
p=0
for(i=1;p<=n;i++)
{
    p=p+i;
}
```

Time Complexity

i	p:	statements
1	0+1	1
2	1+2	1
3	1+2+3	1
4	1+2+3+4	1
5	1+2+3+4+5	1
6	1+2+3+4+5+6	1
k	1+2+3+4+....+k	???

$= 1+2+3+4+....+k > n$

$= k(k+1)/2 > n$

$= k^2 + k/2 > n$

$\cong k^2 > n$

$k <= \sqrt{n}$

$\therefore O(\sqrt{n})$

③
```
for(i=1 ; i<n; i=i*2)
{
    Statements;
}
```

Time Complexity

i	Statements
$1*2^0$	1
$1*2$	1
$1*2*2$	1
$1*2*2*2$	1
...	...
...	...
2^k	1

$i > n$

$i > 2^k$

$2^k >= n$

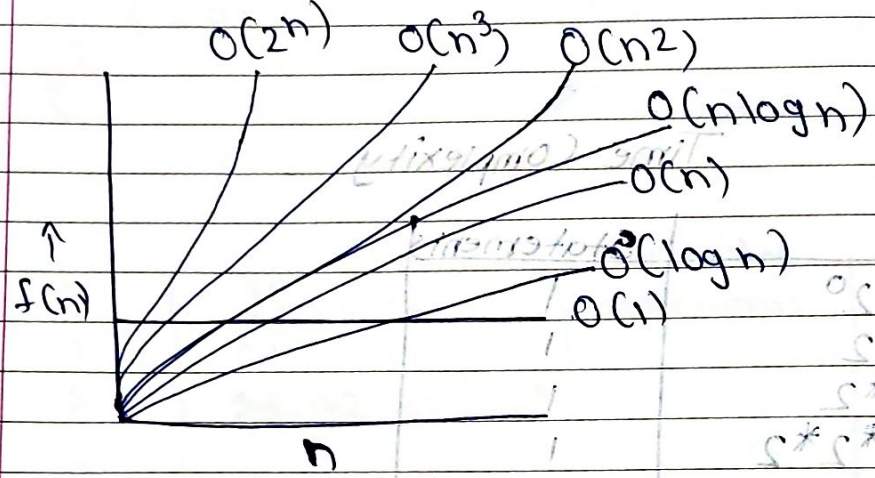
$2^k = n$

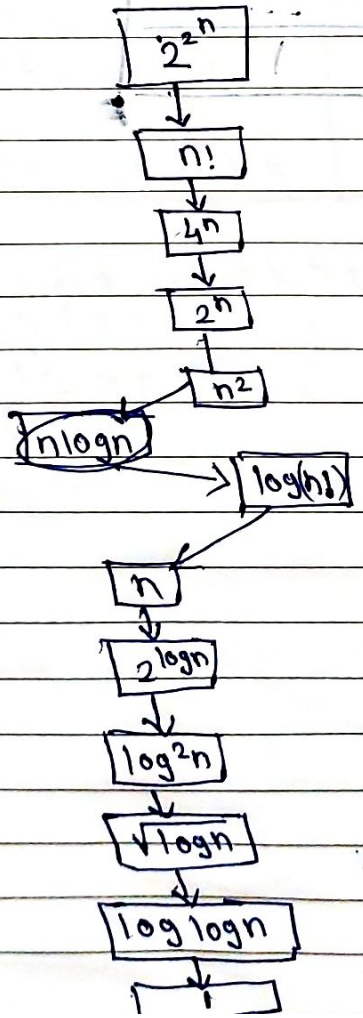
$k = \log_2 n$

$= O(\log_2 n)$

* Rate of Growth

Rate at which the running time increases as a function of input





Decreasing Rates of growth.

Big - OH (O)

~~Big - Omega (~~

① $f(n) = 3n + 8$

$g(n) = 4n$ $\quad c = 4$

$f(n) \leq cg(n)$

$3n + 8 \leq 4n$

$\therefore n = 6$

$= 3(6) + 8 \leq 4(6)$

$= 26 \leq 24$

---

② $f(n) = 3n + 8$

$g(n) = 5n$

$3n + 8 = 5n$

$8 = 2n$

$n = 4 = n_0$

---

③ $f(n) = 3n + 8$

$g(n) = 5n$

~~$f(n) = n^2 + 1$~~

~~$g(n) = 2n^2$~~

~~n_0~~

(4) $f(n) = n^2 + 1$

$g(n) = 2n^2$

$f(n) \leq g(n)$

$n^2 + 1 \leq 2n^2$

$1 \leq n^2$

$n \geq \sqrt{1}$

$n \geq 1$

$n_0 = 1.$

---

(5) $f(n) = \frac{3}{2} n^2 + 1$

let $g(n) = 2n^2$

$\frac{3n^2}{2} + 1 = 2n^2$

$1 = 2n^2 - \frac{3}{2} n^2$

$1 = \frac{1}{2} n^2$

$n^2 = 2$

$n = \sqrt{2}$

Big Omega (Ω)

$f(n) = 5n^2$

$cg(n) \leq f(n)$

$5n^2 \leq 5n^2$

$n = 1 = n_0$

$f(n) = \Omega(g(n))$

Big Theta (Θ)

$f(n) = 4n^2 + 4n + 1$

$g(n) = n^2$

$4n^2 \leq 4n^2 + 4n + 1 \leq 5n^2$ $\quad \therefore n = 1$

$9 \neq 5$

$\therefore n = 2$

$16 \leq 25 \leq 20$

$n = 3$

$36 \leq 49 \leq 45$

Substitution Method

Two - type of substitution
a) forward - Best to n case.
b) Backward

~~Backward~~

Forward Substitution (Factorial)

Q $T(n) = T(n-1) + c \quad n > 1$
$= d \quad n = 1$

i) $T(1) = d$

ii) $T(2) = T(2-1) + c$
$= d + c$

iii) $T(3) = T(3-1) + c$
$= T(2) + c$
$= d + 2c$

iv) $T(n-2) = T(n-2-1) + c$
$= T(n-3) + c$
$= d + T(n-3)c$

v) $T(n) = d + (n-1)c$
$= O(n)$

Backward Substitution (Factorial)

Q $T(n) = T(n-1) + n \quad n > 1$ ——— (1)
$= 1 \quad n = 1$

$T(n-1) = T(n-1-1) + n - 1$
$= T(n-2) + (n-1)$
$T(n) = T(n-2) + n + (n-1)$ ——— (2)

$T(n-2) = T(n-2-1) + n-2$

$T(n-3) + n-2$

$T(n) = T(n-3) + (n-2) + (n-1) + n$ ——— (3)

It will go till k^{th} iteration

$T(n) = T(n-k) + (n-(k-1)) + (n-(k-2))$

$n-k=1$ $\quad$ $n-(n-1-1)$

$k = n-1$ $\quad$ 2

$T(n) = T(1) + (n-1-n+1) + (n-(n-1-1=2)) + \dots$

$= T(1) + 2 + 3 + 4 + \dots + n$

$= 1 + 2 + 3 + \dots + n$

$= n(n+1)$

$T(n) = \Theta^2(n^2)$

$$T(n) = n + 2T\left(\frac{n}{2}\right) \quad \begin{matrix} n>1 \\ n=1 \end{matrix}$$

$$T\left(\frac{n}{2}\right) = \frac{n}{2} + 2T\left(\frac{n}{4}\right)$$

$$T(n) = n + 2\left[2T\left(\frac{n}{4}\right) + \frac{n}{2}\right]$$

$$= n + 4T\left(\frac{n}{4}\right) + n$$

$$= 4T\left(\frac{n}{4}\right) + 2n \quad \text{or} \quad 2^2 T\left(\frac{n}{2^2}\right) + 2n$$

$$= 2^{\frac{2}{6}} T\left(\frac{n}{4}\right) = \frac{n}{4} + 2T\left(\frac{n}{8}\right)$$

$$T(n) = 4\left[\frac{n}{4} + 2T\left(\frac{n}{8}\right)\right] + 2n$$

$$= n + 8T\left(\frac{n}{8}\right) + 2n$$

$$= 2^3 T\left(\frac{n}{2^3}\right) + 3n$$

$$T(n) = 2^k T\left(\frac{n}{2^k}\right) + kn$$

$$\frac{n}{2^k} = 1 \Rightarrow k = \log n$$

$$= nT(1) + n\log n$$

$$= n + n\log n$$

$$= \Theta(n\log n)$$

$T(n) = a\,T(n-b) + f(n)$

$f(n) = O(n^k)$

1) $a < 1$

$T(n) = O(n^k)$

2) $a = 1$

$T(n) = O(n^{k+1})$

3) $a > 1$

$T(n) = O(n^k\, a^{n/b})$

---

① $T(n) = T(n-1) + c$

$\therefore f(n) = O(1)$, $\boxed{\therefore n^k = 1 \quad k = 0}$

$T(n) = O(n^{0+1})$

$= O(n)$

---

② $T(n) = T(n-1) + n$

$T(n) = O(n^2)$

$\therefore a = 2 > 1$

$n^k = n^1$

$k = 1$

$T(n) = O(n^k\, a^{n/b})$

$= O(n \cdot 2^n)$

~~$= O(n)$~~

Page No:
Date:

③ $T(n) = 3T(n-1)$ $n>0$ $a = 3 > 1$
$= c$ $n=0$ $n^k = n^1$
$k = 1$

~~$a = 3 \geq 1$~~
~~$n^k = n^1$~~

$f(n) = 0$
$\therefore k = 0$

$T(n) = O(n^k \cdot a^{n/b})$
$= O(3^{n/1})$ ~~$= O(3^1)$~~
$= O(3^n)$

---

④ $T(n) = T(n-1) + n(n-1)$ if $n \geq 2$
$= 1$ ⟶ $n = 2$

$f(n) = O(n^2)$
$n^k = n^2$
$k = 2$

~~$T(n) = O(2$~~
$T(n) = O(n^{2+1})$
$= O(n^3)$

---

⑤ $T(n) = T(n-1) + \log n$ ⟶ (1)

$T(n-1) = T(n-2) + \log(n-1)$
$T(n) = T(n-2) + \log(n-1) + \log n$ ⟶ (2)
$T(n-2) = T(n-3) + \log(n-2)$
$T(n) = T(n-3) + \log(n-2) + \log(n-1) + \log n$
⋮
k^{th} $T(n) = T(n-k) + \log n + \log(n-1) + \log(n-2)$
$\cdots + \log(n-k+1)$

$n-k=0$
$n=k$

$\therefore \log a + \log b = \log(ab)$

$T(n) = T(n-1) + \log(n) + \log(n-1) + \log(n-2) + \ldots + \log(1)$
$= 1 + \log(n \cdot (n-1)(n-2) \ldots 3 \cdot 2 \cdot 1)$
$= 1 + \log(n!)$

$\therefore T(n) = O(\log n^n)$
$= O(n \log n)$

$T(n) = T(\sqrt{n}) + \log n$ ——— (1)

Let $n = 2^m$
$m = \log_2 n$

$T(2^m) = T(2^{m/2}) + m$ ——— (2)
$S(m) = T(2^m)$
$S(m) = 2S\left(\frac{m}{2}\right) + m$ ——— (3)

~~$S(m) = 2S\left(\frac{m}{2}\right) + m$~~

$a=2, b=2, k=1, p=0$

$S(m) = O(m^k \log^{p+1} m)$
$S(m) = O(m \cdot \log m)$

~~$T(n)$~~
$S(m) = T(2^m)$
$n = 2^m$
$m = \log n$

$$T(n) = \Theta(\log n \cdot \log\log n)$$

Min-Max

$$T(n) = 2T\left(\frac{n}{2}\right) + 2 \qquad \because n > 2$$

$$n = 2$$

$$= 2$$

$$\Theta(n \cdot \log n)$$

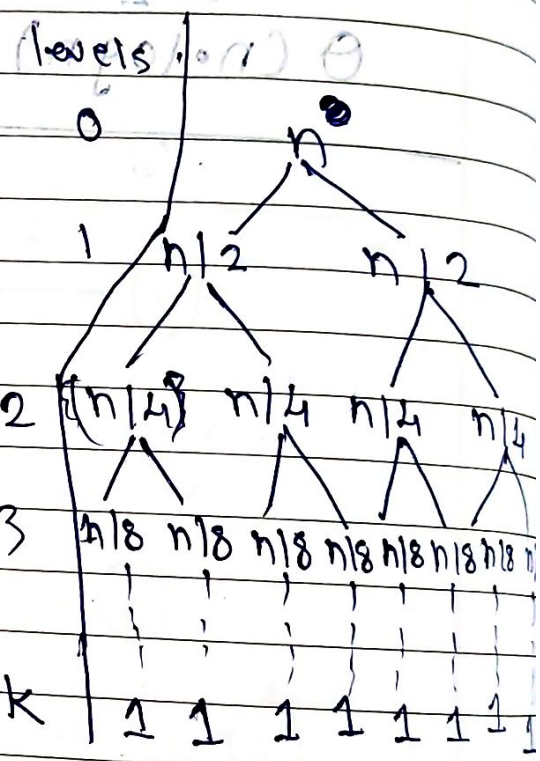
Page No.:
Date:

Recurrence Tree Method:

$T(n) = aT\left(\frac{n}{b}\right) + f(n)$

- a → No. of subproblems.
- $\frac{n}{b}$ → size of subproblem
- $f(n)$ ← cost incurred for dividing & combining

$T(n) = 2T\left(\frac{n}{2}\right) + n$



$\frac{n}{2^k} = 1$

$n = 2^k \Rightarrow k = \log n$

no. of nodes.	cost
1	n
2	n
4	n
8	n
⋮	⋮
$n = 2^k$	n
	n^k

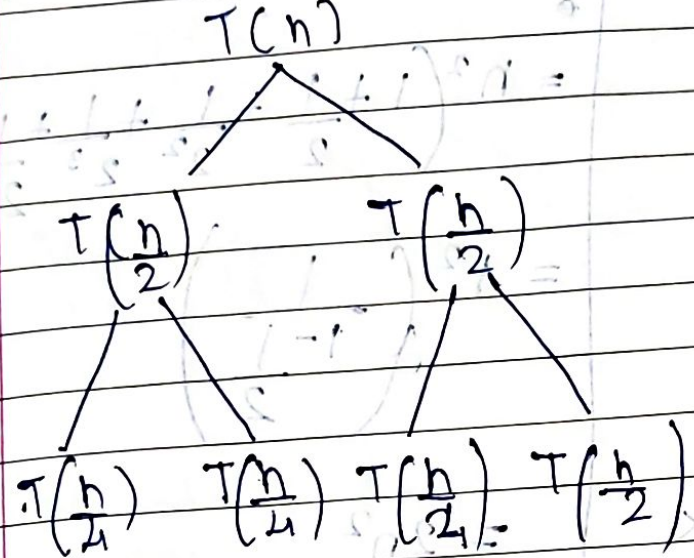
pos = 2

Total Cost = n^k

$= n \times \log n$

$T(n) = \Theta(n \log n)$

$T(n) = 2T\left(\frac{n}{2}\right) + n^2$



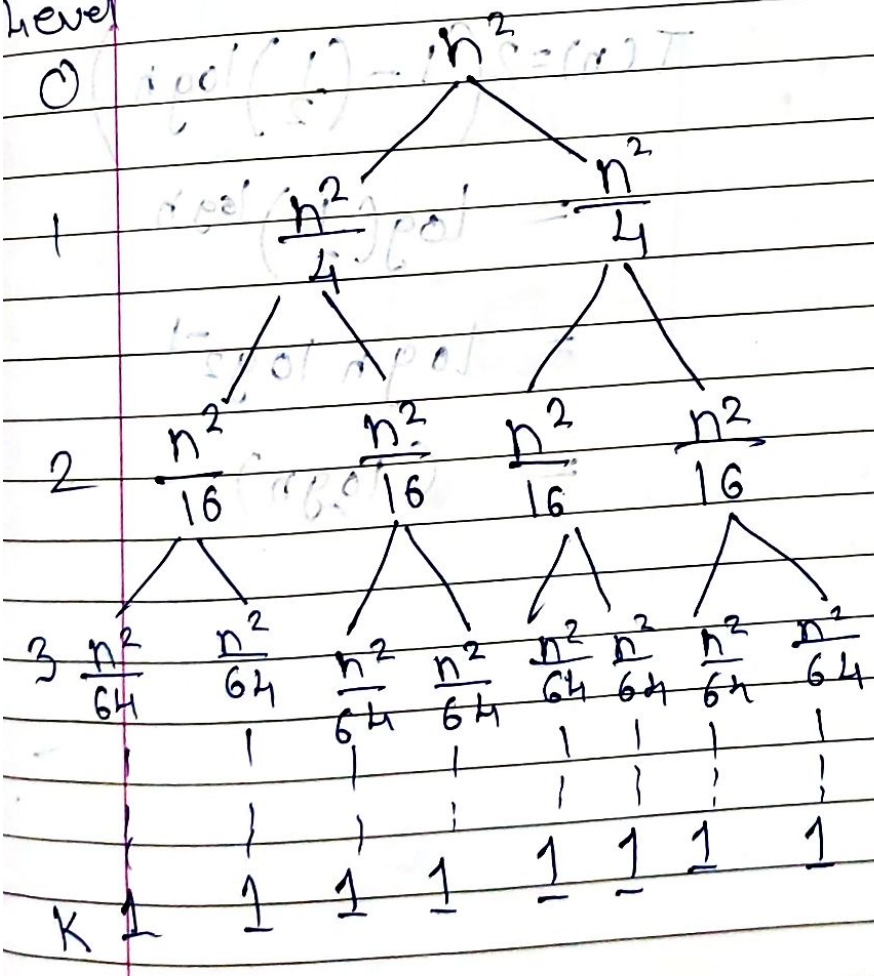
$T(n) = 2T\left(\frac{n}{2}\right) + n^2$

$T\left(\frac{n}{2}\right) = 2T\left(\frac{n}{4}\right) + \left(\frac{n}{2}\right)^2$

$T\left(\frac{n}{4}\right) = 2T\left(\frac{n}{8}\right) + \left(\frac{n}{4}\right)^2$

$T\left(\frac{n}{8}\right) = 2T\left(\frac{n}{16}\right) + \left(\frac{n}{8}\right)^2$

Level



no. of nodes.	cost
1	n^2
2	$n^2/2$
4	$n^2/4$
	$n^2/8$
$n=2^k$	$n^2/2k$

Total cost $= n^2 + \frac{n^2}{2} + \frac{n^2}{4} + \frac{n^2}{8} \cdots$

$= n^2\left(1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} \cdots\right)$

$= 1^2\left(\frac{1}{1-\frac{1}{2}}\right)$

$= 2n^2$

$T(n) = \Theta(n^2)$

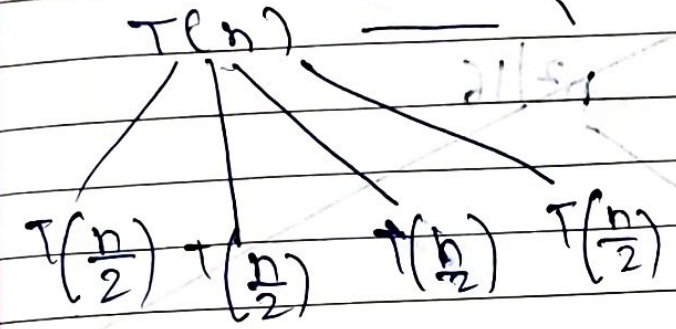
$T(n) = 2\left(1 - \left(\frac{1}{2}\right)\log n\right)$

$= \log\left(\frac{1}{2}\right)^{\log n}$

$= \log n \log 2^{-1}$

$= 1\ (\log n)$

$$T(n) = 4T\left(\frac{n}{2}\right) + n^2$$



no. of nodes	Cost	Total Cost = kn^2
1	n^2	$T(n) = \Theta(n^2 \log n)$
4	$n^2/2$	
16	$n^2/4$	
4^k	~~$k n^2$~~ kn^2	
	~~$\log n$~~	

$$\frac{n}{2^k} = 1$$

$$n = 2^k$$

$$k = \log n$$

$T(n) = 3T\left(\frac{n}{4}\right) + n^2$ $\quad n = 1$ $\quad n > 1$

Level $\quad T\left(\frac{n}{4}\right) \quad n^2/16$

$T\left(\frac{n}{4^2}\right) \quad \frac{n^2}{16^2}$

~~$T(n)$~~

$T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + n \quad n > 1$

Level 0 $\quad T(n)$ $\quad$ maximum depth.

1 $\quad T\left(\frac{n}{3}\right) \quad T\left(\frac{2n}{3}\right)$

$T\left(\frac{n}{3^2}\right) \quad T\left(\frac{2n}{3^2}\right) \quad T\left(\frac{2n}{3^2}\right) \quad T\left(\frac{2^2 n}{3^2}\right)$

$T\left(\frac{n}{3^3}\right) \quad T\left(\frac{2n}{3^3}\right) \quad T\left(\frac{2n}{3^3}\right) \quad T\left(\frac{2^2 n}{3^3}\right) \quad T\left(\frac{2n}{3^3}\right) \quad T\left(\frac{2^2 n}{3^3}\right) \quad T\left(\frac{2^2 n}{3^3}\right) \quad T\left(\frac{2^3 n}{3^3}\right)$

$T(1) \quad T(1) \quad T(1) \quad T(1) \quad T(1) \quad T(1) \quad T(1) \quad T(1)$

k^{th} $\quad T\left(\frac{2^k}{3^k} n\right)$

$T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + n$

$T\left(\frac{n}{3}\right) = T\left(\frac{n}{3^2}\right) + T\left(\frac{2n}{3^2}\right) + \frac{n}{3}$

$T\left(\frac{2n}{3}\right) = T\left(\frac{2n}{3^2}\right) + T\left(\frac{2^2 n}{3^2}\right) + \frac{2n}{3}$

$T\left(\frac{n}{3^2}\right) = T\left(\frac{n}{3^3}\right) + T\left(\frac{2n}{3^3}\right) + \frac{n}{3^2}$

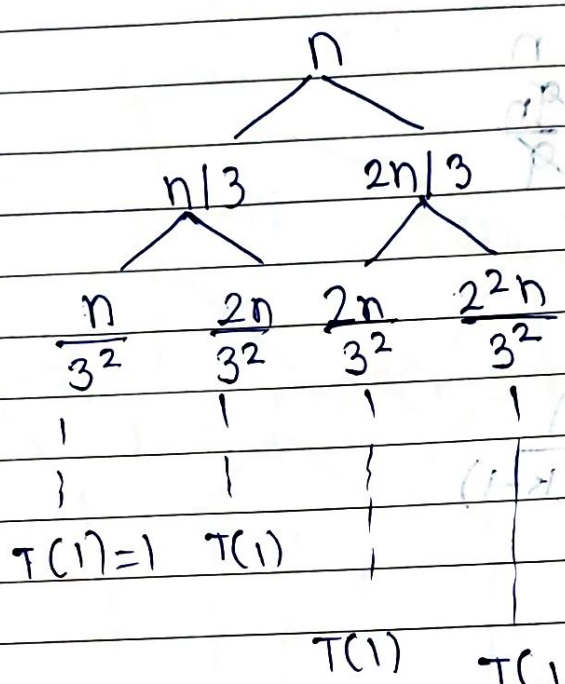
$T\left(\frac{2n}{3^2}\right) = T\left(\frac{2n}{3^3}\right) + T\left(\frac{2^2 n}{3^3}\right) + \frac{2n}{3^2}$

$T\left(\frac{2^2 n}{3^2}\right) = T\left(\frac{2^2 n}{3^3}\right) + T\left(\frac{2^3 n}{3^3}\right) + \frac{2^2 n}{3^2}$

Page No.:
Date:

No. of nodes.

Cost

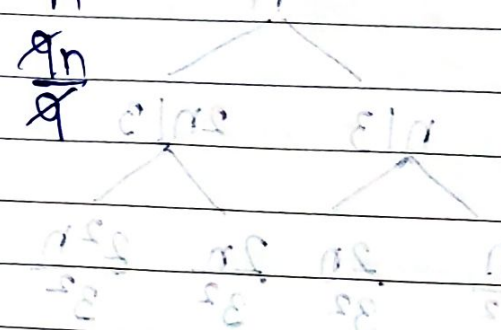


$$T\left(\left(\frac{2}{3}\right)^k n\right) = T(1)$$

$$\frac{2^k}{3^k}\, n = 1$$

$$n = \left(\frac{3}{2}\right)^k \Rightarrow k = \log_{3/2} n$$

no. of nodes	cost
1	n
2	n
4	~~4n~~
~~4~~	
⋮	⋮
2^{k-1}	$\frac{n}{n(k-1)}$



$T(n) = L_c + N_c$

$= 2^k + n(k-1)$

~~$= 2^{\log_{3/2} n} +$~~

$= 2\log_{3/2}^{n} + n\log_{3/2}^{n}$

$= n\log_{3/2}^{2} + n\log_{3/2}^{n}$

$\leq O(n \log_2 n)$

$$T(n) = 2T(n-1) + 1 \quad n > 0$$
$$n = 0$$

Level

0 $T(n)$

1 2 $T(n-1)$ $T(n-1)$

2 $T(n-2)$ $T(n-2)$ $T(n-2)$ $T(n-2)$ 2

3 $T(n-3)$ ...

k $T(n-k)$ $T(0)$ $T(0)$ $T(0)$ $T(0)$ $T(0)$

$n-k=0$

$n=k$

~~Total cost~~

Cost

1

2

4

8

⋮

2^k

$T(n) = 1+2+4+\cdots+2^k$

$T(n) = \frac{a(r^{n+1}-1)}{r-1}$

$= 1\left(\frac{2^{n+1}-1}{1}\right)$

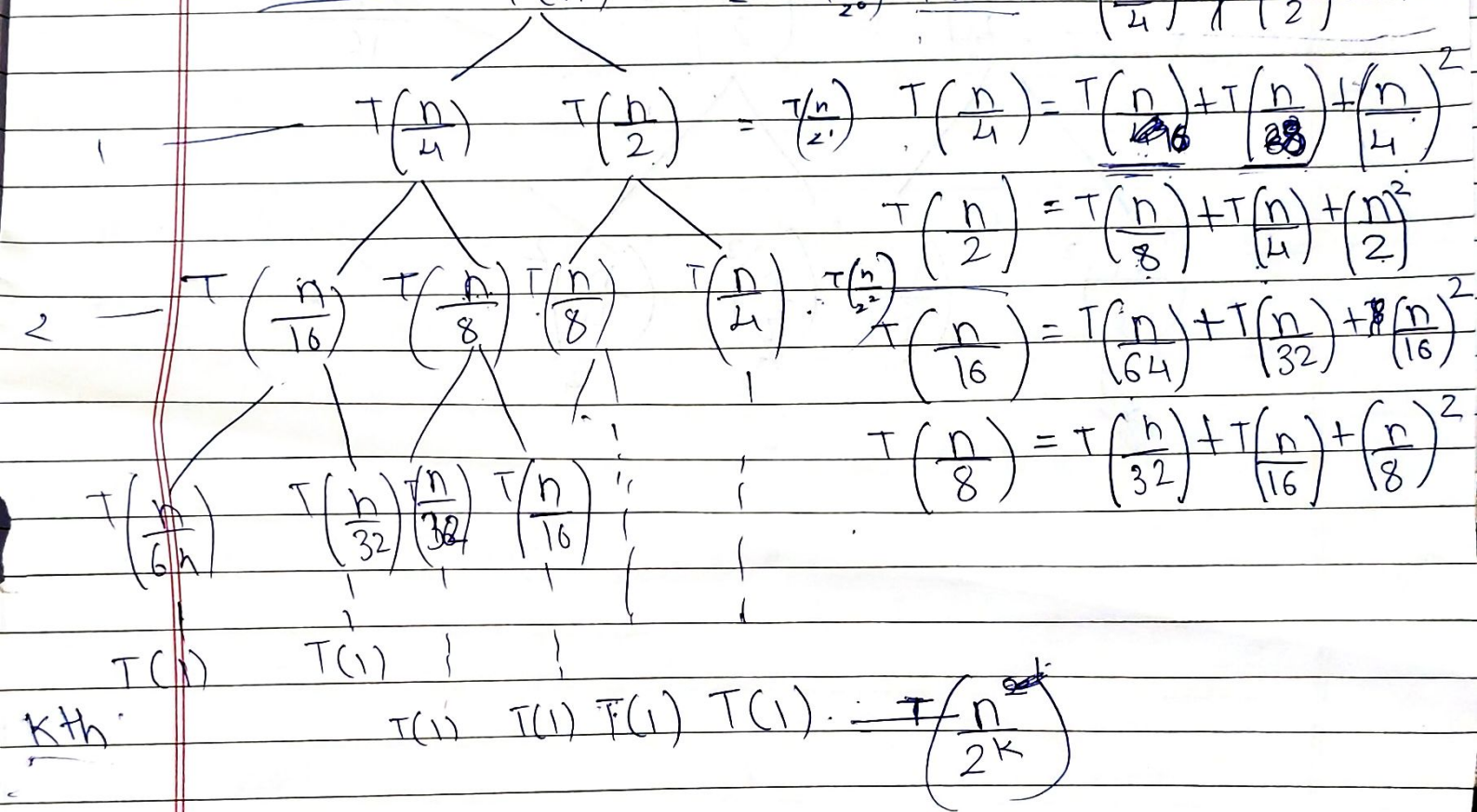
$T(n) = O(2^n)$

$T(n) = 1+2+4+8+16+\cdots 2^k$

$f(x) = x^2 + 3x + 8$

$f(3) = ?$ $\quad f(x/2) = ?$

$$T(n) = T\left(\frac{n}{4}\right) + T\left(\frac{n}{2}\right) + n^2 \quad n>1$$
$$n=1$$



$\therefore T(1) = T\left(\frac{n}{2^k}\right)$ — Find k

$1 = \frac{n}{2^k}$

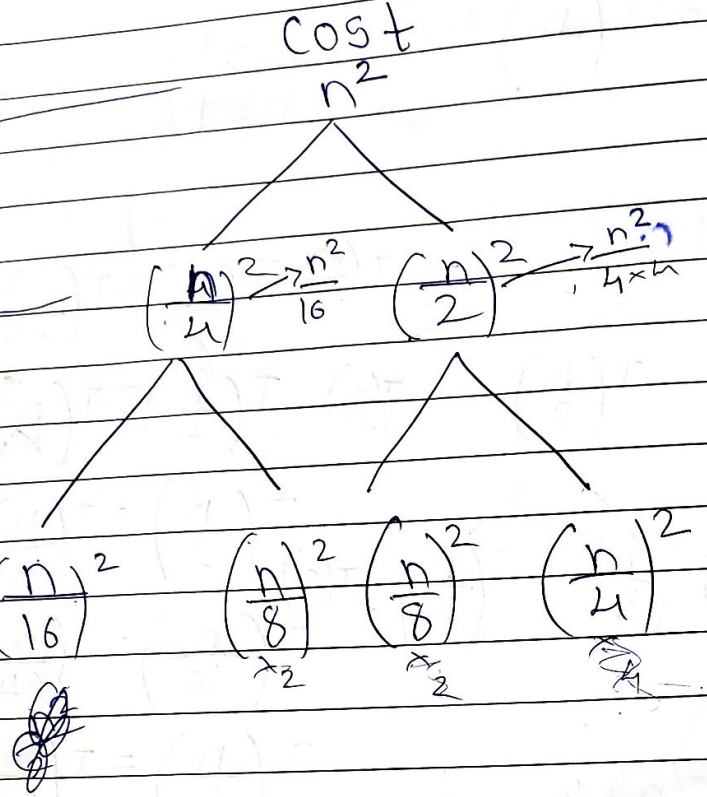
$2^k \times 1 = n$

$n = 2^k$

$2^3 = 8$

$\therefore \log_2 8 = 3$

$$\log_2 n = k$$

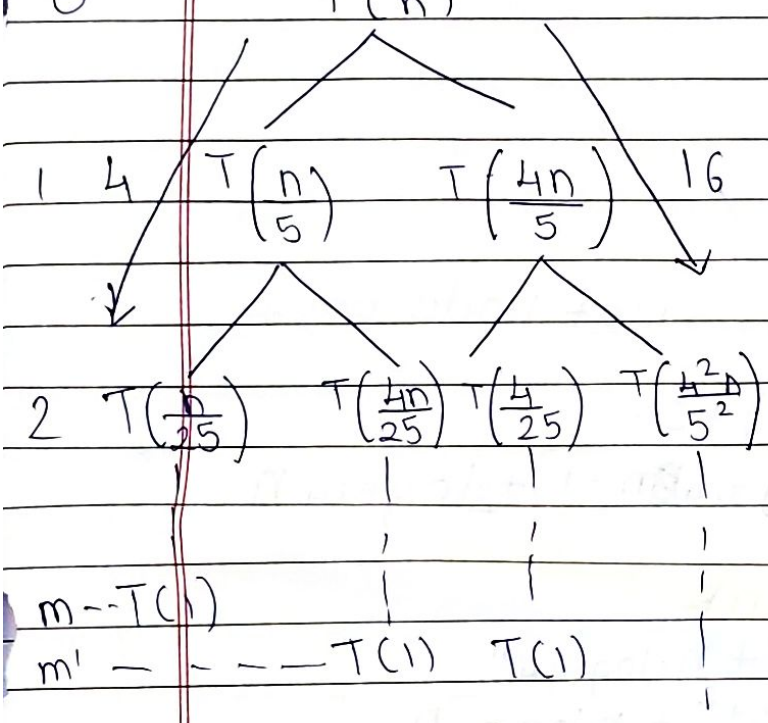


n^2

$\frac{5n^2}{16}$

$$T(n) = T\left(\frac{n}{5}\right) + T\left(\frac{4n}{5}\right) + n \rightarrow \text{Cost}$$

Level 0



$$T(n) = T\left(\frac{n}{5}\right) + T\left(\frac{4n}{5}\right) + n$$

$$T\left(\frac{n}{5}\right) = T\left(\frac{n}{5^2}\right) + T\left(\frac{4n}{5^2}\right) + \frac{n}{5}$$

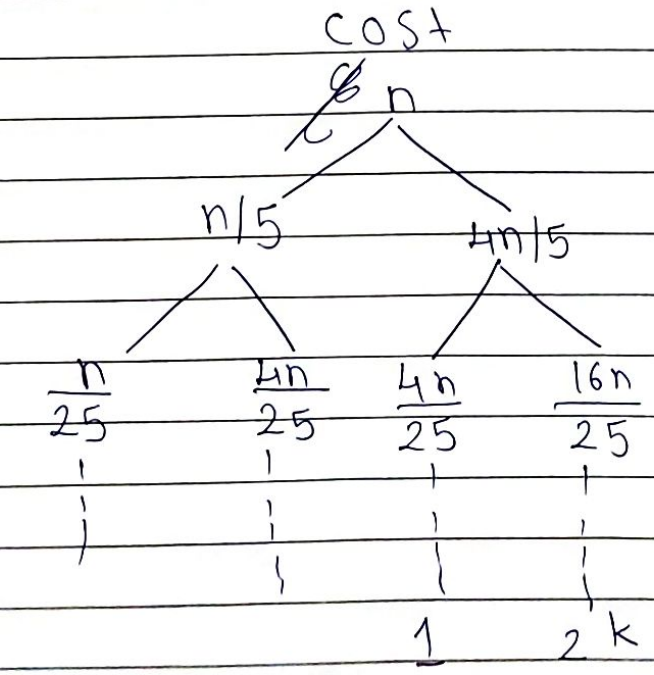
$$T\left(\frac{4n}{5}\right) = T\left(\frac{4n}{5^2}\right) + T\left(\frac{4^2 n}{5^2}\right) + \frac{4n}{5}$$

$$T\left(\frac{n}{25}\right) = T\left(\frac{n}{5^3}\right) + T\left(\frac{4n}{5^3}\right) + \frac{n}{25}$$

$$T\left(\frac{4n}{25}\right) = T\left(\frac{4n}{5^3}\right) + T\left(\frac{4^2 n}{5^3}\right) + \frac{4n}{5^2}$$

$$T\left(\frac{16}{25} n\right) = T\left(\frac{4^2 n}{5^3}\right) + T\left(\frac{4^3 n}{5^3}\right) + \frac{4^2 n}{5^2}$$

~~Cost~~



k^{th} term $= \left(\frac{4}{5}\right)^k n$

$1 = \left(\frac{4}{5}\right)^k n$

$n = \left(\frac{5}{4}\right)^k$

$k = \log_{5/4} n$ → leaf node cost

Total cost $= n(k-1) + 2^k$

$= n(\log_{5/4} n - 1) + 2^{\log_{5/4} n}$

↓

Total cost $= 2^k + nk$

$= 2^{\log_{5/4} n} + n \cdot \log_{5/4} n$

$= n^{\log_{5/4} 2} + n \log_{5/4} n$

$= O(n^3)$